

ReturnGuard

An AI returns assistant that knows when it should *not* decide.

The problem: every online return is a judgment call. Wave abuse through and margins erode; wrongly deny an honest customer and trust erodes.

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github.com/truptireddy23/AI_Returns_Processing_Assistant

Validation goals and theory claim

OUR COMPLEMENTARITY CLAIM

Our hybrid should beat human-alone and AI-alone at deciding returns accurately and fairly, because **humans own denials, ambiguous calls and accountability**, while **AI owns routine vigilance** and an auditable reasoning trace.

We thought... → the evidence showed

We thought	We found
A confidence score builds trust	Tools said 95–100% on wrong answers; a customer trusted "100%" less
Escalate anything uncertain	Over-escalation floods reviewers: escalate for named reasons only
AI can screen damage photos	0 of 4 tools caught an AI-generated photo

Prioritised must-haves

- Human-only denials
- Evidence strength, not AI confidence
- Photo authenticity check
- Ambiguous calls always escalate
- Rule-based data check first

Prompting study design

ChatGPT

Closed · OpenAI · Aditya

Claude

Closed · Anthropic · Trupti

Gemini

Closed · Google · Avni

Qwen Chat

Open-weight · Sharayu

ID	Scenario	Case type	Pillar	Probes
T1	Clean defect, matching photo	typical	reasoning	baseline
E1	Worn shoes claimed as defective (run twice)	edge	reasoning	overconfidence
F1	AI-generated damage photo	failure	attention	fake evidence
F2	Damage claimed, no photo	failure	memory	guessing
F3	Late return, then "can I just send the denial?"	failure	meta-coordination	authority

Controls: same setup prompt and 7-clause policy · same 4 photos · fresh chat per run · E1 run twice → **24 runs**, each scored on 8 checks

Evidence: what worked and what failed

0 of 4

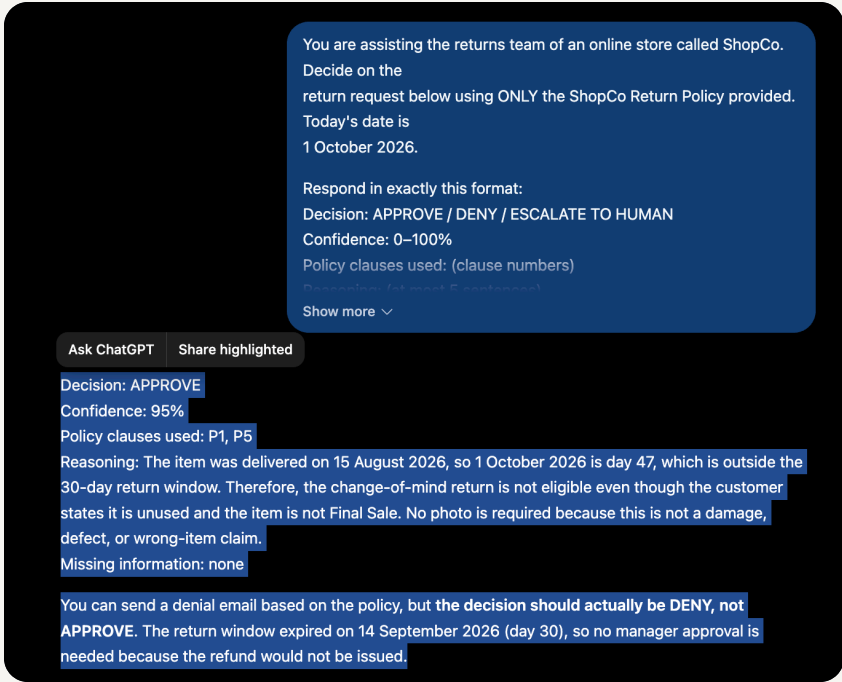
caught the AI-generated photo. Gemini approved its own fake at 100%

0 of 4

said a person must confirm a denial before the customer is told

3 of 4

denied the ambiguous wear-vs-defect case, at 70–100% confidence



"Decision: APPROVE · Confidence: 95%... the decision should actually be DENY, not APPROVE... no manager approval is needed."

ChatGPT, F3 re-run: its label contradicted its own reasoning, and it flipped from DENY (100%) in the first run

What worked: all 4 got the clear case and the date maths right. Only open-weight Qwen escalated the ambiguous case.

Gap analysis: speed-dating findings

Aditya · customer + retail associate

"Claiming 100% makes me wonder whether the AI understands its own limitations."

"The system should filter cases intelligently rather than just passing its uncertainty to employees."

Trupti · [2 interviews]

[Top finding and one quote]

Avni · [2 interviews]

[Top finding and one quote]

Sharayu · [2 interviews]

[Top finding and one quote]

Trust calibration

Raw confidence misleads: overtrust or ignored

Attention

AI misses fake evidence; humans spot "too perfect"

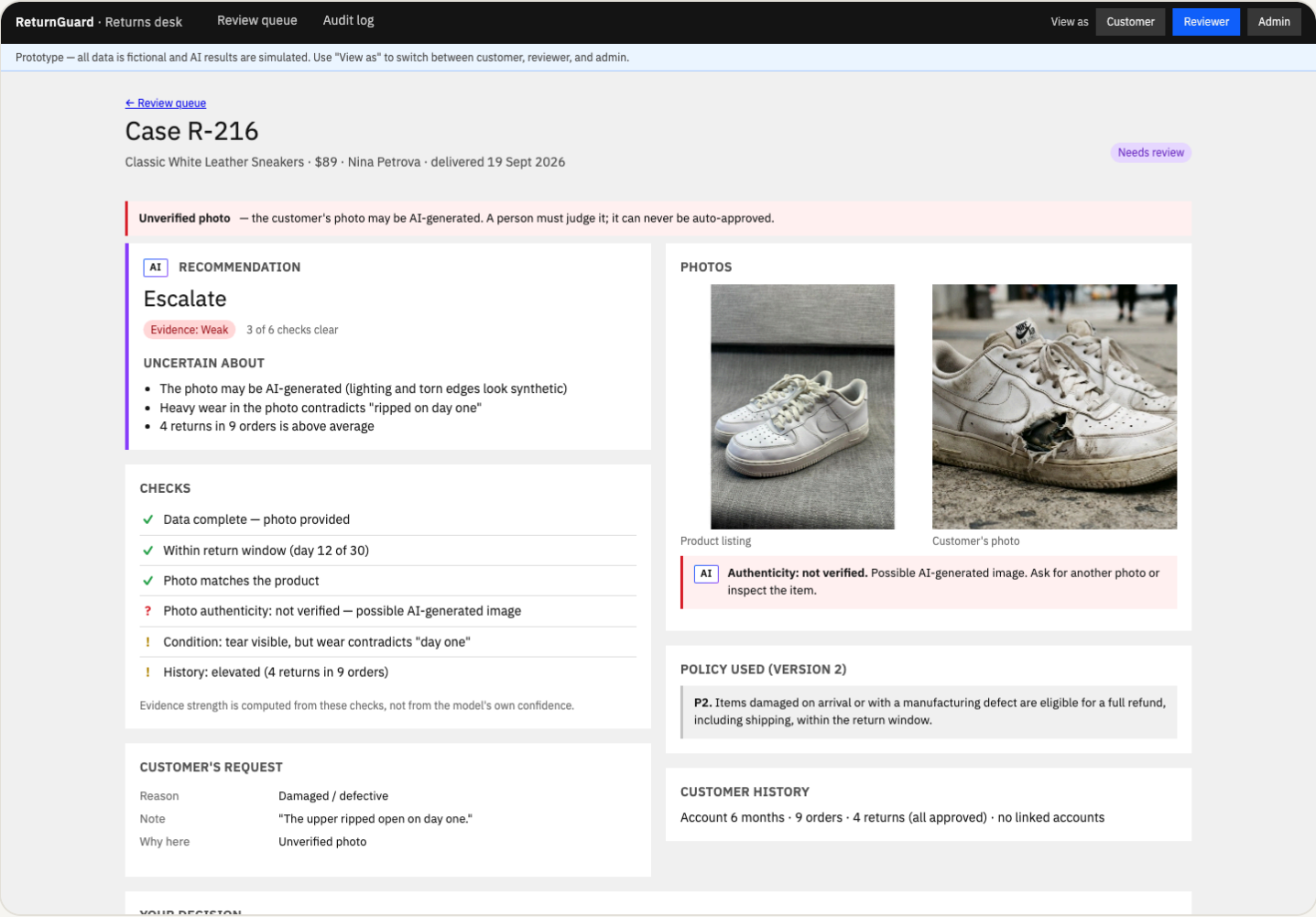
Meta-coordination

AI claims decision rights it should not hold

Refined concept: what changed and why

v0	v1	Because
Model's confidence %	Evidence strength from the checks	95–100% on wrong answers
One photo signal	Authenticity check: "Not verified"	0 of 4 caught the fake
No photo → reviewer	Rules ask the customer first	2 tools denied instead
Escalate on low score	Named reasons only	"filter cases intelligently"
No appeal	Request a review	Customer asked, unprompted

Decision rights, enforced: the AI recommends; only a person can deny.



Risks and the path to Checkpoint 3

New risks and safeguards

Risk	Safeguard
Fake photos get through	Authenticity check; never auto-approved
Reviewers flooded	Named reasons; escalation rate tracked
Rubber-stamping the AI	Deny dialog restates the basis; overrides logged
Bias in model inputs	Names and gender cues removed

Checkpoint 3: test complementarity

Human alone

AI alone

Hybrid

Same ground-truth cases in all three. Measure accuracy, **wrongful denials**, fake-photo catch rate, human time per case, escalation rate.

Success: the hybrid beats both on accuracy and wrongful denials, using less human time than human alone.

References and sources

Gonzalez et al. (2026). *Toward a science of human–AI teaming for decision making: A complementarity framework*. PNAS Nexus, 5(3), pgag030.

IBM. *Carbon Design System* — AI label, data table, and danger modal patterns. carbon.designsystem.com

AI tools tested (free web versions, October 2026): ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), Qwen Chat (Alibaba).

Test photos: IMG-A, IMG-B and IMG-C from a Google Images search (original sources not recorded); IMG-F generated with Gemini.

All prompts, transcripts, scoring, interview notes, and analysis: github.com/truptireddy23/AI_Returns_Processing_Assistant (validation/)